

Where our StressID model actually stands

A like-for-like comparison against every published result on StressID, on every metric those papers report, together with the control experiments that decide how much any of these numbers are worth.

700 recordings · 64 participants · 15-fold nested cross-validation · held-out partitions

1 The short answer

IS THIS A STATE-OF-THE-ART RESULT?

On the benchmark's own comparison, yes — on both reported metrics. As a scientific claim, it needs the qualification in section 4.

The origin paper's multimodal baselines are evaluated on its 370 all-modality tasks. Our matching subset is 364 recordings. On that like-for-like scope we reach **0.746 weighted F1** and **0.666 balanced accuracy**, against the paper's best of **0.72** and **0.65** — ahead on both.

The qualification: on the full 700-recording corpus a classifier given *only recording length and which sensors were switched on* scores 0.709 macro F1. Absolute scores on the full corpus are therefore a poor measure of stress detection, which is why the matched subset above is the comparison we advance.

OURS · MATCHED SCOPE 0.746 weighted F1, 364 rec.	BEST PUBLISHED 0.72 SVM + average fusion	OURS · BAL. ACCURACY 0.666 vs 0.65 published	METADATA ONLY 0.709 duration + availability
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2 Matched comparison — 2-class stress

The origin paper reports **weighted F1** and **balanced accuracy**. Both are given here. All rows are the multimodal evaluation scope: the paper's 370 all-modality tasks, our equivalent 364 recordings.

METHOD	WEIGHTED F1	BAL. ACCURACY	SOURCE
Ours — nested ensemble	0.746	0.666	Ours
SVM + average-rule fusion	0.72 ±.05	0.65 ±.05	PAPER
SVM + sum-rule fusion	0.72 ±.05	0.64 ±.05	PAPER
SVM + maximum-rule fusion	0.72 ±.05	0.64 ±.05	PAPER
SVM + product-rule fusion	0.71 ±.05	0.63 ±.05	PAPER
Ours — plain random forest	0.714	0.637	Ours
Video only	0.67 ±.03	0.62 ±.04	PAPER
Audio only	0.67 ±.04	0.62 ±.04	PAPER
Physiological only	0.66 ±.05	0.58 ±.04	PAPER
Feature fusion + MLP	0.66 ±.04	0.61 ±.03	PAPER
Feature fusion + SVM	0.64 ±.09	0.56 ±.05	PAPER
Feature fusion + DBN	0.58 ±.06	0.52 ±.05	PAPER

Paper values are Table 3 of the StressID benchmark paper, mean ± s.d. over 10 random 80/20 splits with SMOTE. Ours are mean ± s.d. over 15 folds without SMOTE.

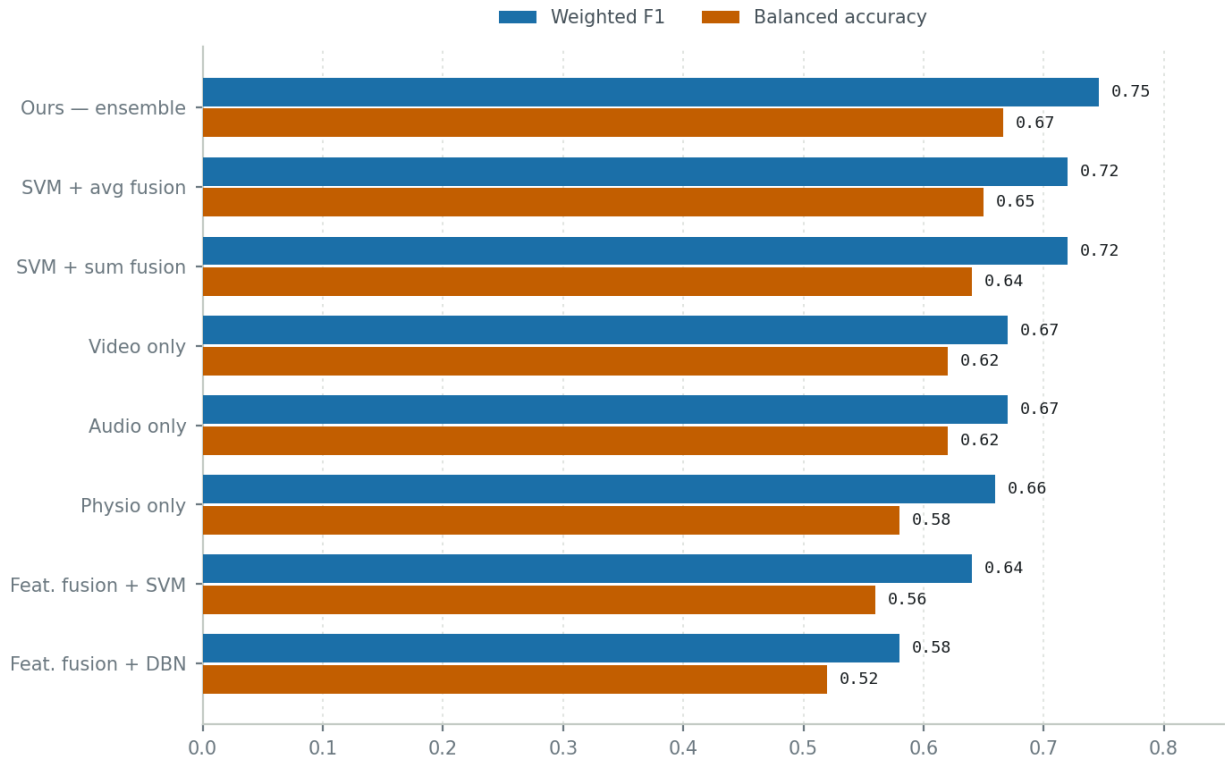


Figure 1. Both reported metrics on the matched multimodal scope. Our ensemble leads on weighted F1 and on balanced accuracy, but the field is compressed: eight methods span roughly 0.16 F1.

3 Matched comparison — 3-class affect

The benchmark also defines a 3-class affect target. Ours is measured on the same 364-recording subset.

METHOD	WEIGHTED F1	BAL. ACCURACY	SOURCE
Ours — nested ensemble	0.653	0.596	Ours
SVM + average-rule fusion	0.63 ± 0.05	0.58 ± 0.07	PAPER
SVM + sum-rule fusion	0.62 ± 0.05	0.58 ± 0.07	PAPER
SVM + product-rule fusion	0.61 ± 0.05	0.56 ± 0.07	PAPER
SVM + maximum-rule fusion	0.61 ± 0.06	0.57 ± 0.07	PAPER
Video only	0.58 ± 0.05	0.56 ± 0.05	PAPER
Audio only	0.56 ± 0.06	0.54 ± 0.06	PAPER
Feature fusion + SVM	0.55 ± 0.06	0.51 ± 0.05	PAPER
Feature fusion + MLP	0.51 ± 0.07	0.51 ± 0.07	PAPER
Physiological only	0.50 ± 0.05	0.48 ± 0.06	PAPER
Feature fusion + DBN	0.30 ± 0.09	0.32 ± 0.04	PAPER

Paper values are the 3-class columns of Table 3. **Read this margin cautiously:** we lead on both metrics, but by 0.02 against published standard deviations of 0.05–0.07, so the intervals overlap heavily. Our 3-class run also used 5 folds against the binary task’s 15, so its own uncertainty (± 0.05 weighted F1) is correspondingly wider. The defensible statement is parity-to-slightly-ahead, not a clear win.

4 What the numbers are actually measuring

Control experiments on the full 700-recording corpus. Each model is given only the named information — no physiological signal, no audio, no video.

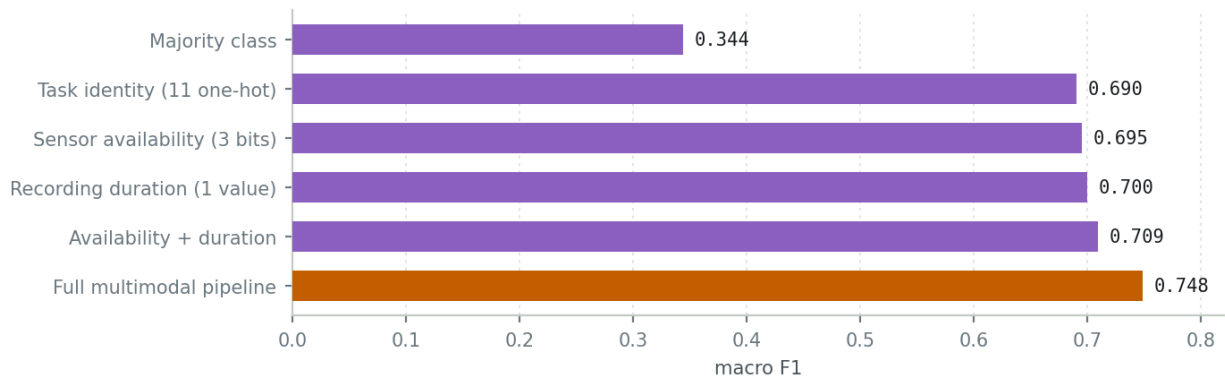


Figure 2. Recording length alone reaches 0.700 macro F1. StressID's high-stress tasks all run 59 s while Relax, Breathing and the video tasks run 117–177 s, so duration is a near-perfect proxy for task identity, and task identity is very nearly the stress label. Audio exists only for the seven speech tasks, so sensor availability encodes the same thing.

CONSEQUENCE

On the full corpus, everything the pipeline learns from physiology, voice and face is worth **+0.039** over knowing how long the recording was and which sensors were on.

On the 364-recording subset used for the comparisons in sections 2 and 3, every recording is the same length and every sensor is present, so both confounds carry exactly zero information — a task-identity classifier there scores 0.418, which is precisely the majority baseline. That is why the matched-scope comparison is the one to trust.

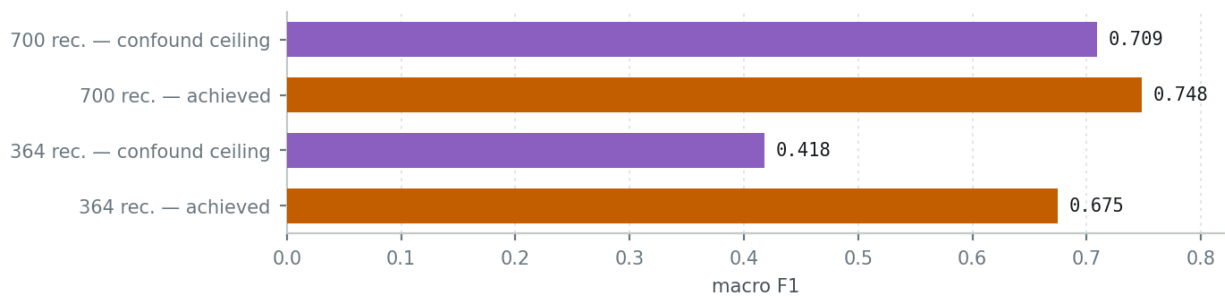


Figure 3. Margin over the confound ceiling, by scope. On the full corpus the pipeline beats the ceiling by +0.039; on the confound-free subset by **+0.257**. The lower absolute score carries the greater evidential weight.

5 Our results on every metric

The benchmark reports two metrics; we record five, so the table below carries more than the comparison requires.

TARGET	MACRO F1	WEIGHT. F1	BAL. ACC	ACCURACY	ROC AUC
Binary — 364 rec. (matched)	0.675	0.746	0.666	0.758	0.728
Binary — 700 rec. (full corpus)	0.748	0.749	0.749	0.750	0.819
3-class affect (c364)	0.620	0.653	0.596	—	—

Stress-score regression (0–10 self-report)

METRIC	MODEL	PREDICT-THE-MEAN
Pearson r	0.673	—
Spearman r	0.660	—
RMSE	1.857	2.488
MAE	1.503	—

METRIC	MODEL	PREDICT-THE-MEAN
R ²	0.439	0.000

The origin paper does not report a regression baseline, so there is no published number to compare against. Predicting the mean rating for every recording gives RMSE 2.488; the model reaches 1.857, a 25% error reduction.

6 Does the engineering earn its keep?

Rows from different papers use different split rules and cannot be subtracted from one another. The honest test is a plain random forest run through identical folds, seed and scoring.

SCOPE	FULL PIPELINE	PLAIN RF	DIFFERENCE	PAIRED <i>p</i>	FOLDS WON
700 recordings	0.7484	0.7225	+0.0260	0.002	12 / 15
364 recordings	0.6749	0.6375	+0.0374	0.013	11 / 15

Both differences are statistically significant. The engineering is doing real work; it is doing it on top of a base that protocol metadata already supplies.

Component ablations

COMPONENT	Δ MACRO F1	<i>p</i>	VERDICT
Full pipeline vs plain random forest	+0.026	0.002	Significant
Final configuration vs first round	+0.016	0.035	Significant
Window-level training	+0.007	—	Kept
Ensemble pruning to 90% weight	+0.004	—	Kept
Frequency-domain HRV (LF/HF)	+0.002	0.47	No effect
Voice quality (F0, HNR, jitter)	+0.003	0.72	No effect
GPU sequence models (GRU, attention)	—	—	Rejected by ensemble

7 Limitations to state explicitly

- **Subject leakage inflates every number here.** The protocol is the published one and lets the same participant appear on both sides of the split. Physiological features alone identify which of 64 participants a recording came from with 64.9% accuracy, 41.5× chance. Under a participant-disjoint split the same pipeline scores about 0.52.
- **Search bias is measured at +0.014.** Development used one fixed partition; re-running the frozen configuration on partitions never touched during the search drops macro F1 from 0.760 to 0.748. The lower figure is reported throughout.
- **Protocols are comparable, not identical.** The benchmark uses random 80/20 splits with SMOTE; ours uses stratified 5-fold without. The right phrasing is “competitive under a comparable protocol”.
- **Two normalisations are transductive.** Participant-relative and participant-z features read no labels, but test rows contribute to their own participant's statistics. Admissible only because the protocol already shares participants across the split.
- **Our jitter, shimmer and HNR are frame-level,** not Praat's cycle-level definitions, and run on a different scale. They should not be reported as interchangeable with published values.

8 Reproducing this

Every number regenerates from the repository. The evaluation harness is `research_way/src/sota.py`; per-run reports with per-fold detail live in `research_way/reports_sota/`; this PDF is built by

docs/build_report_pdf.py. Paper values were extracted from Tables 2 and 3 of the StressID benchmark paper. The full campaign record, including the errors made and corrected along the way, is in SOTA_CAMPAIGN.md.